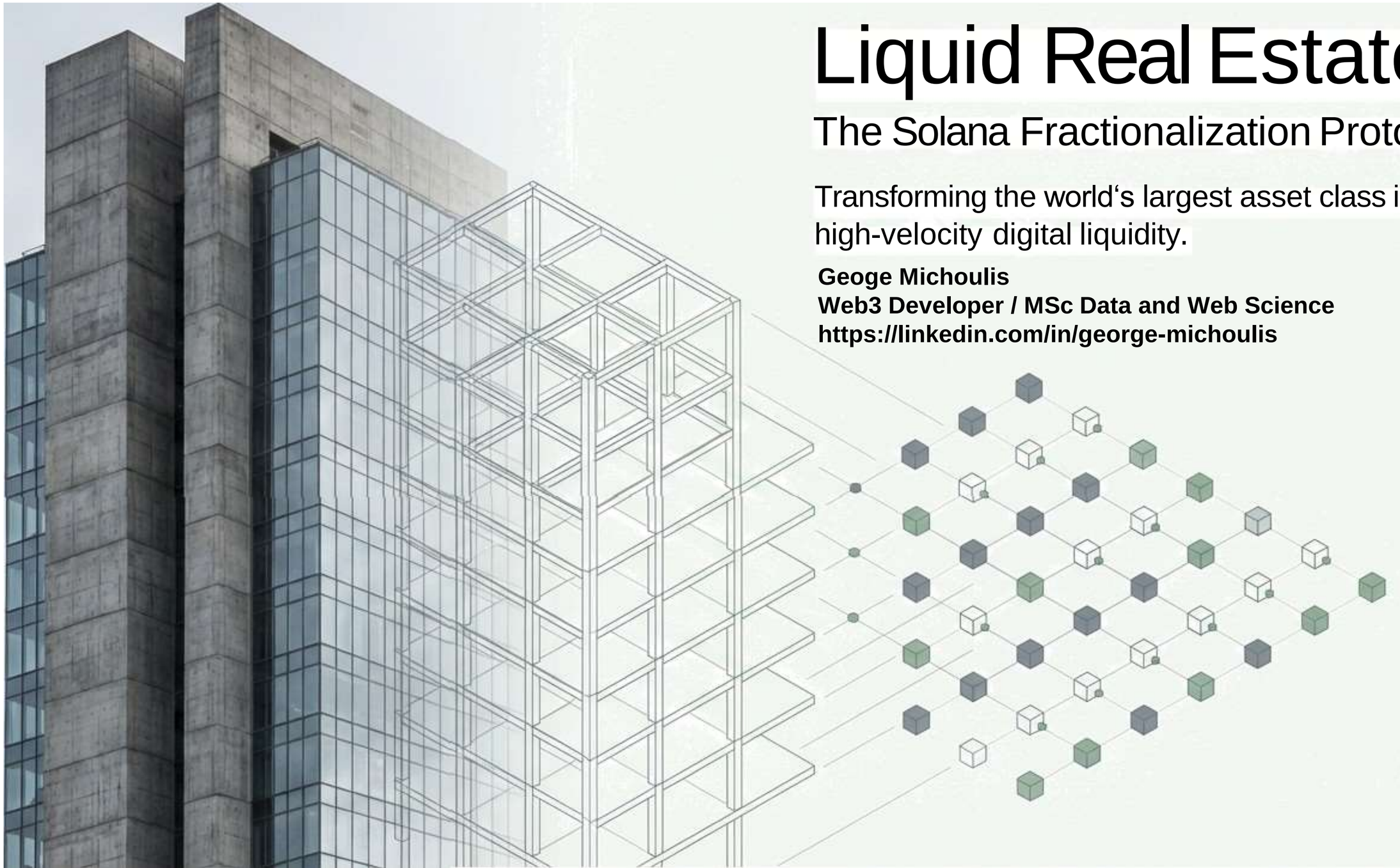
Liquid Real Estate

The Solana Fractionalization Protocol

Transforming the world‘s largest asset class into high-velocity digital liquidity.

Geoge Michoulis
Web3 Developer / MSc Data and Web Science
https://linkedin.com/in/george-michoulis



The $330 Trillion Illiquidity Problem

Exclusionary Capital

Premium commercial investments require $500,000+ minimums, locking out retail and mid-tier participants.

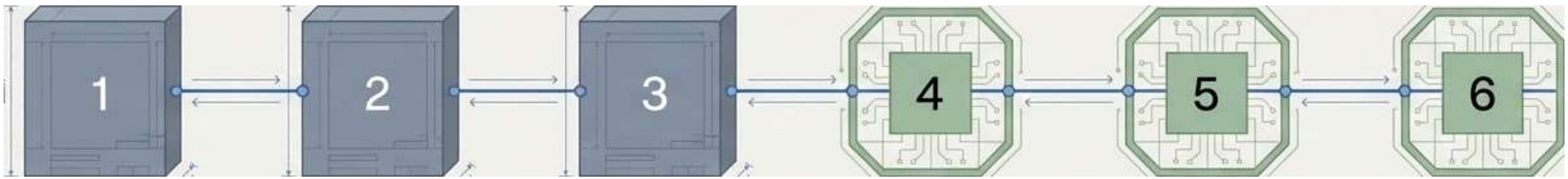
Trapped Value

Severe illiquidity. The average commercial real estate exit takes 6+ months to settle in static structures.

Opaque Intermediaries

Heavy operational friction. Brokers, lawyers, and legacy escrow systems erode yields and obscure real-time valuation.

The Asset-to-Liquidity Pipeline



Physical Layer

Physical Deed undergoes legal audit and valuation.

Legal Layer

Legal SPV holds the title, ensuring bankruptcy remoteness.

Asset Layer

Master NFT minted as a single, immutable digital record.

Escrow Layer

PDA Vault smart contract securely locks the NFT.

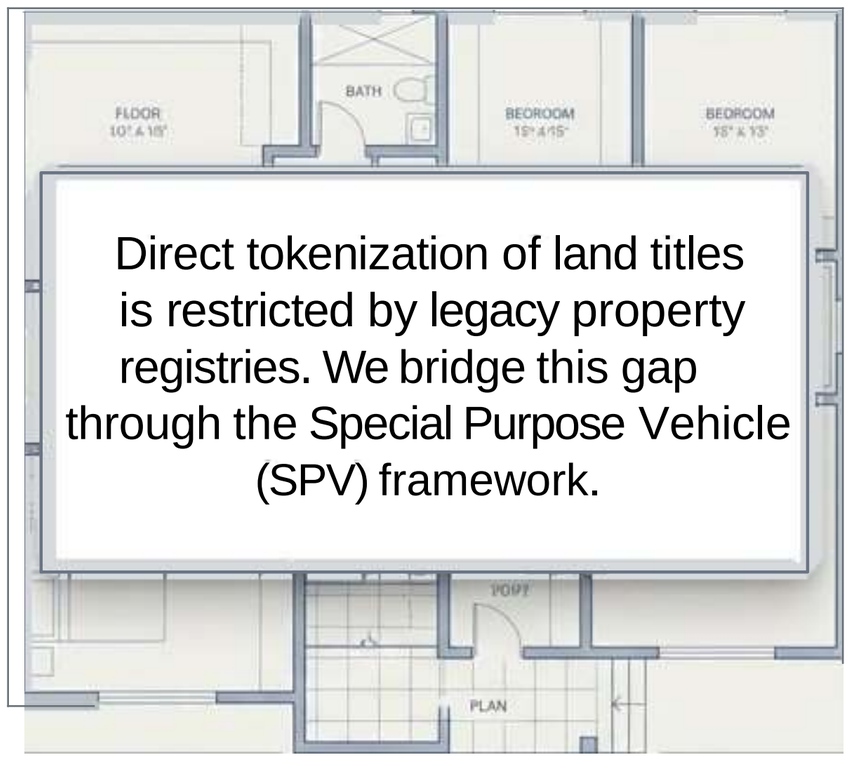
Fractional Layer

SPL Tokens minted as thousands of fractional shares.

Liquidity Layer

DEX Trading opens global 24/7 automated markets.

The Legal Wrapper: Bridging Law to Ledger



Direct tokenization of land titles is restricted by legacy property registries. We bridge this gap through the Special Purpose Vehicle (SPV) framework.

Bankruptcy Remoteness

Property title is held by an independent LLC or Trust, protecting the asset from protocol or developer insolvency.

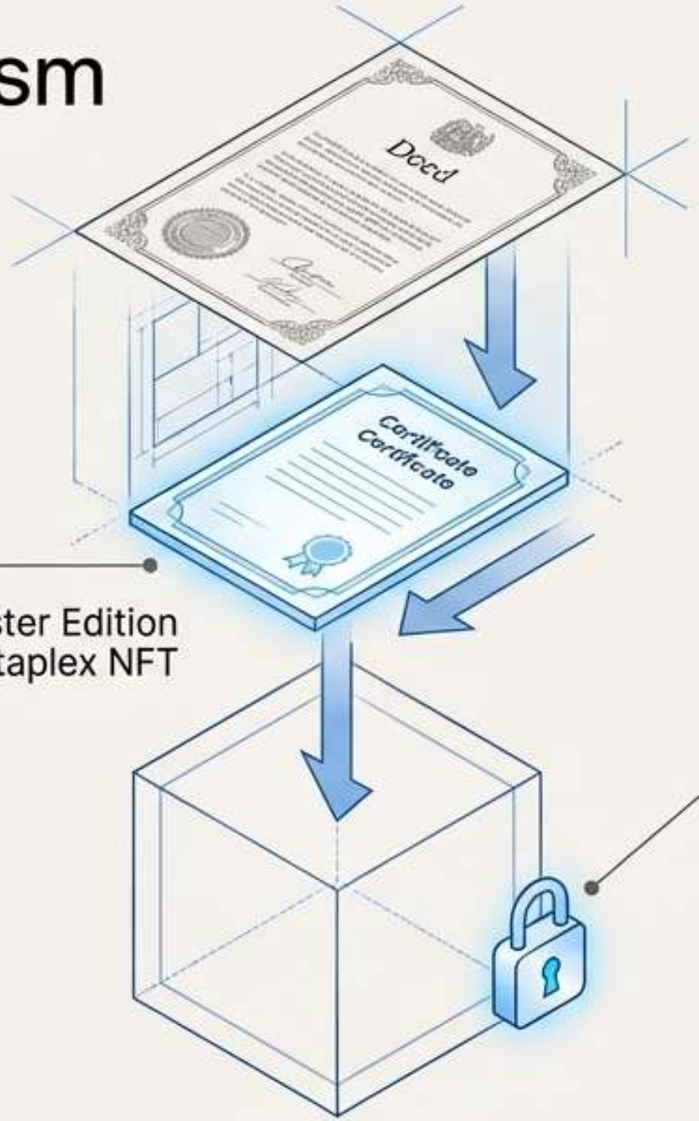
Liability Limitation

Token holders are shielded from personal liability beyond their fractional investment, mirroring traditional stock protections.

Legal Enforceability

The SPV operating agreement legally recognizes the blockchain ledger as the definitive, binding record of ownership.

Trustless Escrow: The PDA Vault Mechanism



The Root Asset

The property is digitized as a 1-of-1 Master Edition NFT containing IPFS-pinned appraisals, title insurance, and legal agreements.

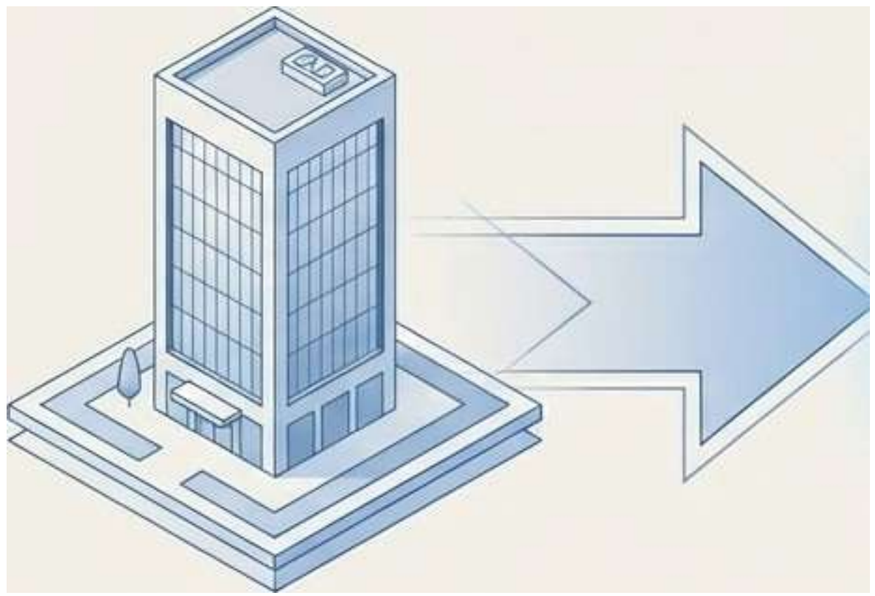
Master Edition
Metaplex NFT

The Lock

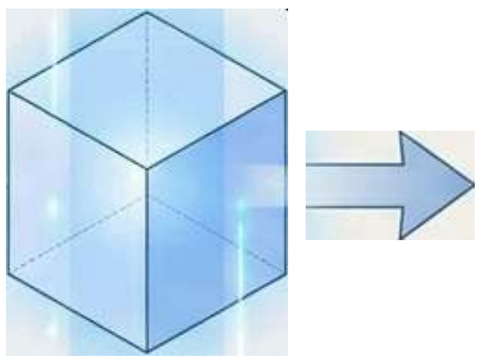
The NFT is deposited into a PDA. Because PDAs have no private keys, the asset cannot be moved by human intervention—only by the pre-programmed consensus of the Anchor smart contract.

Program Derived
Address (PDA) Vault

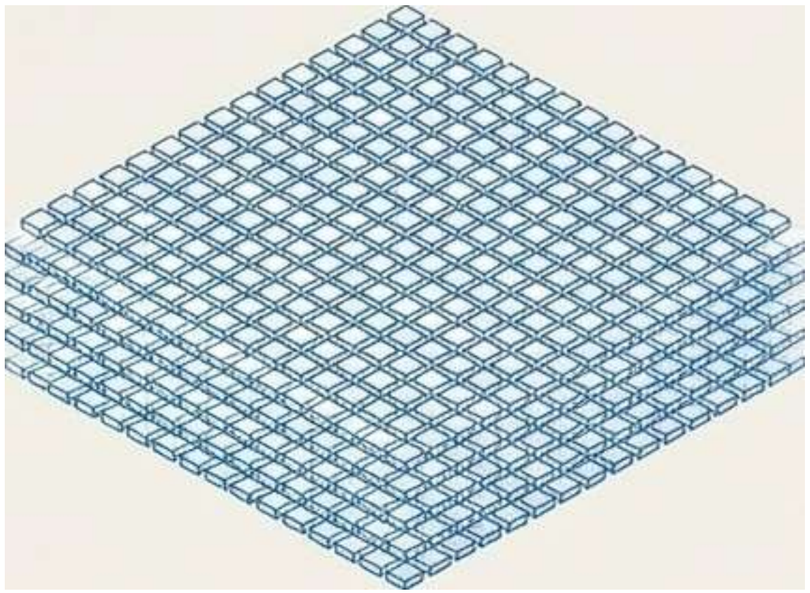
Fractionalization: Democratizing Capital



Value: $5,000,000



Smart Contract Minting



100,000SPLTokens
@ $50 each

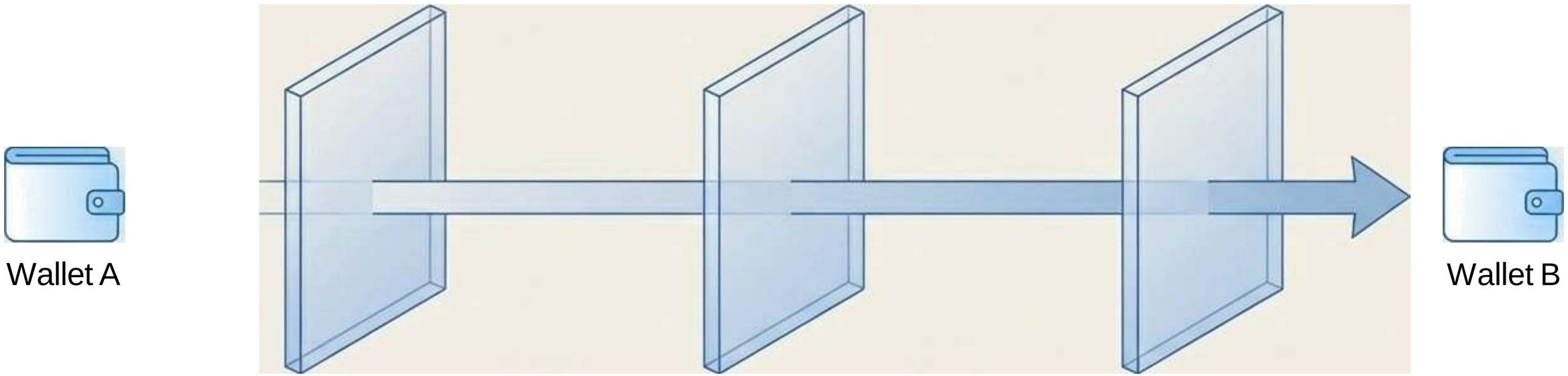
Once the Master NFT is cryptographically locked in the PDA Vault, the contract automatically mints a predefined supply of SPL tokens.

These tokens represent a proportional, economic claim on the underlying asset‘s yield and appreciation.

Minimum investment thresholds drop from institutional levels ($500k+) to micro-retail levels ($50).

Programmable Compliance via Token-2022

Legacy tokens cannot handle institutional compliance. We utilize Solana's Token-2022 extensions to bake regulatory logic directly into the asset.



Transfer Hooks

Automatically checks off-chain KYC/AML registries in real-time before allowing a transfer.

Default Account State

All new token accounts are initialized as Frozen until the user submits compliance data, at which point the backend executes a Thaw instruction.

Permanent Delegate

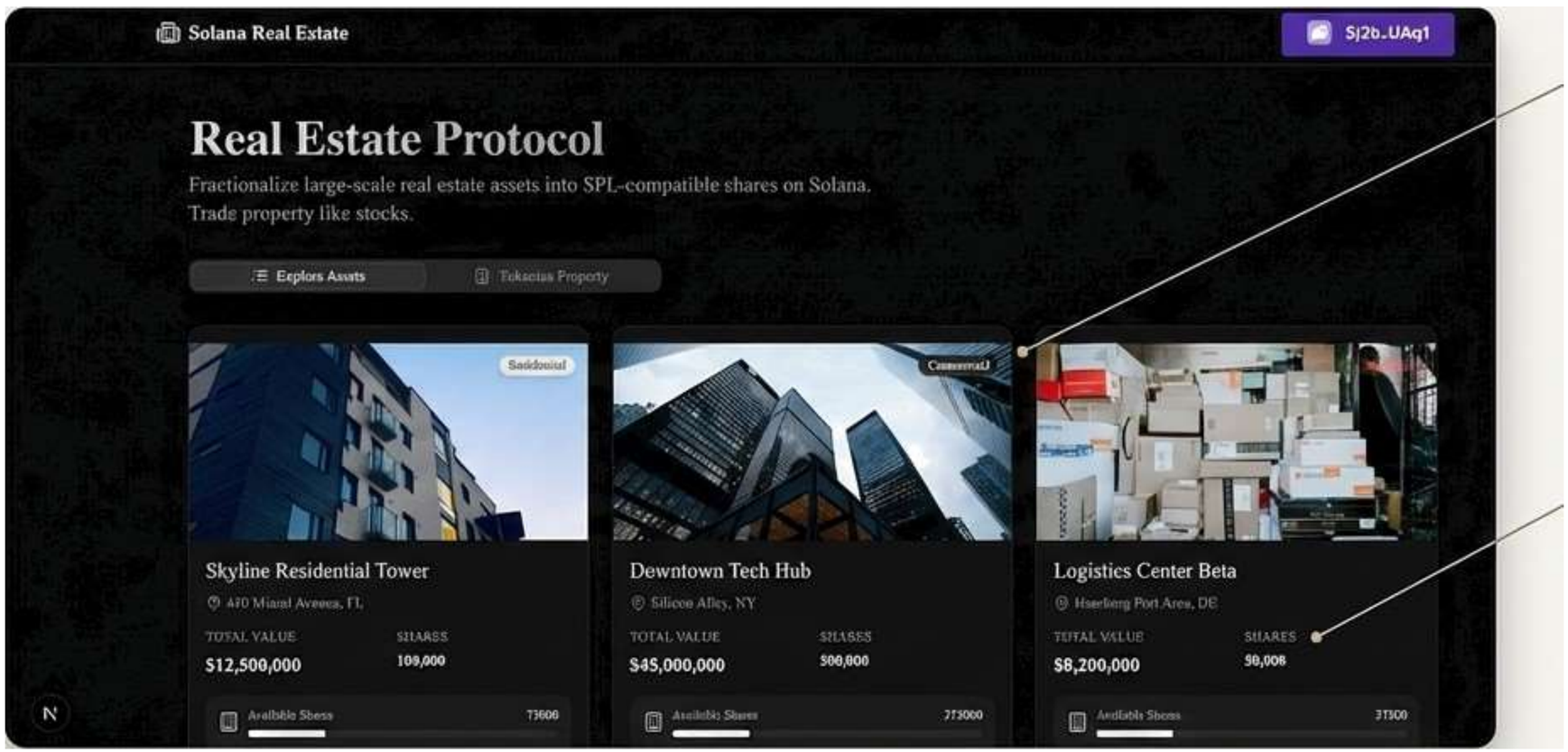
An administrative override key allowing the protocol to recover assets in the event of legal court orders or lost private keys.

The Liquidity Engine: Why Meteora DLMM?

Dimension	Traditional AMM	**Meteora DLMM**
Liquidity Model	Full-range Static Curve	Discrete Price Bins
Fee Structure	Fixed 0.25%	Dynamic (0.01%–1%) based on volatility
Capital Efficiency	Low; capital sits idle out of range	Ultra-High; captures volume surges effectively

Takeaway: **Meteora's dynamic fees compensate Liquidity Providers during high-volatility price discovery events**, making it **the perfect engine for RWA tokens.**

The Front-End:
Trading Property Like Stocks



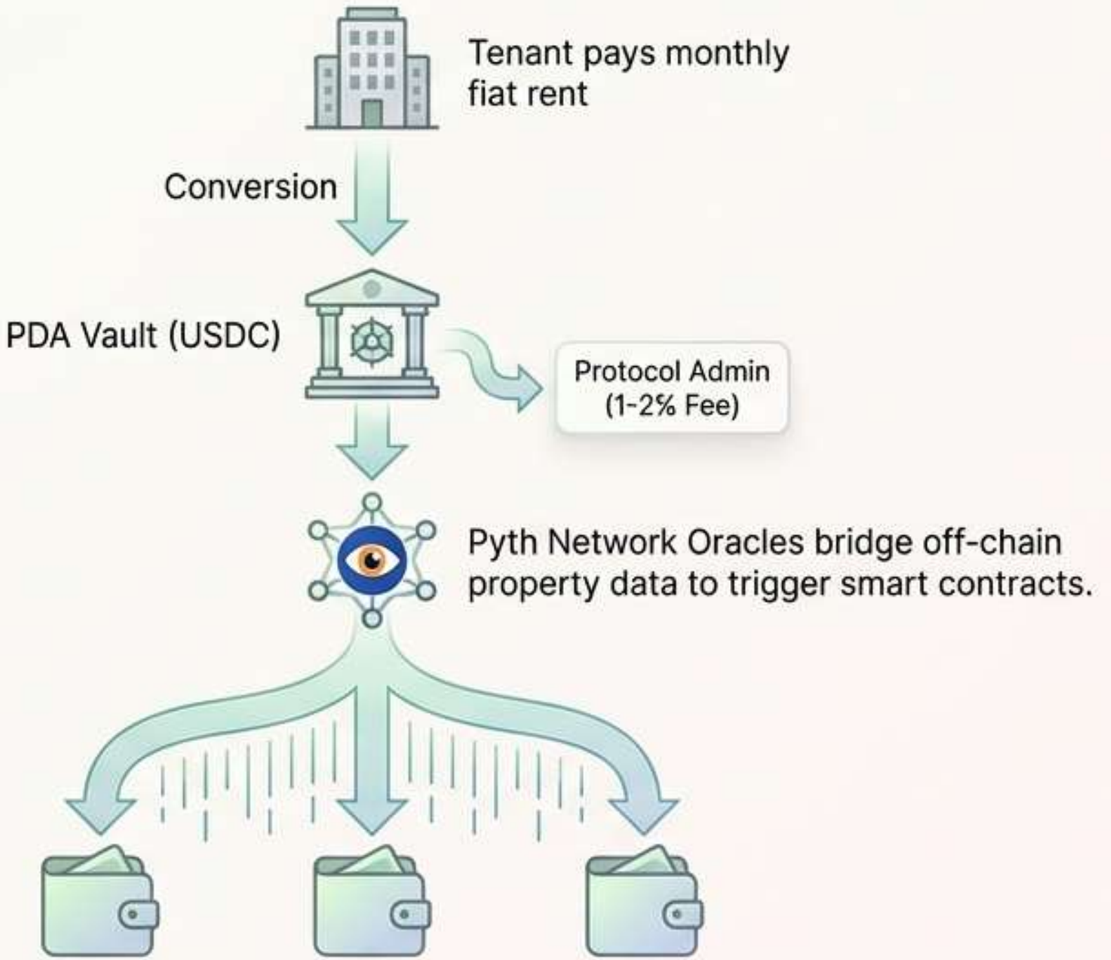
Curated Premium Assets

Institutional-grade commercial, residential, and industrial properties audited and pre-vetted.

Micro-Fractionalization

Total property values broken down into highly liquid, SPL-compatible shares accessible to global wallets.

The Yield Waterfall: Automated Dividend Distribution

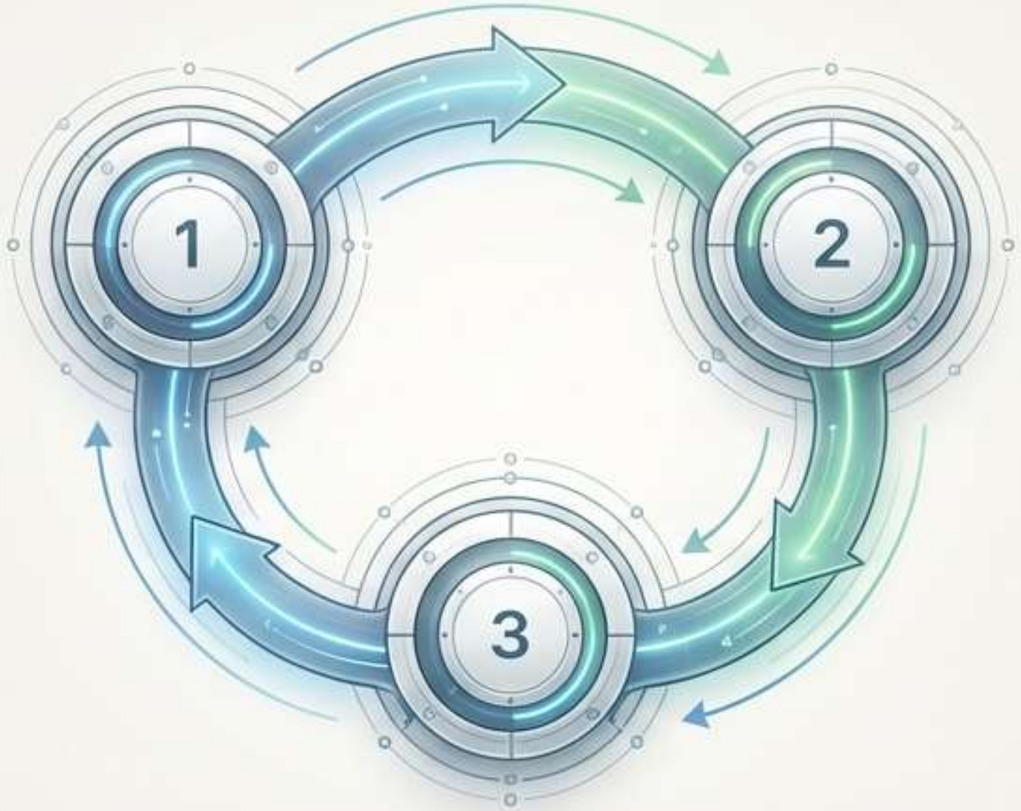


Smart contract automatically distributes remaining USDC yield pro-rata to all fractional token holders instantly.

Business Strategy: Multi-Stage Value Capture

Origination Fees

1-2% flat fee charged to real estate developers for legal auditing, SPV formation, and minting the asset on-chain.



Marketplace Trading Fees

0.1% swap fee captured by the protocol on every secondary market transaction moving through the frontend order book.

Yield Administration

1-2% recurring management fee skimmed from generated rental income before USDC distribution to token holders.

NotebookLM

The Evolution of Real Estate: Protocol vs. REITs

	Traditional REITs	Solana Protocol
Trading Hours	Market Hours (9:30 - 4:00)	24/7/365 Global Markets
Settlement Speed	T+1 or T+2 Days	~400 Milliseconds Atomic Settlement
Transparency	Quarterly Opaque Reports	Real-time On-chain Verification
Minimum Investment	$500+ broker minimums	$50 micro-transactions
Governance	Centralized Fund Managers	Smart Contract & DAO Voting

Target Audience & Global Use Cases

Retail Investors

Everyday Wealth Generation

Enables retail users to gain exposure to high-yield real estate without requiring massive down-payments, mortgages, or pristine credit scores.

Real Estate Developers

Accelerated Crowdfunding

Allows builders to tokenize finished properties for fast capital recuperation, bypassing expensive, high-interest traditional bank loans.

Global Portfolio Allocators

Borderless Wealth Protection

Investors in economically volatile regions can instantly protect their wealth by buying fractions of stable, high-yield US/EU real estate via self-custody wallets.

The Solana Imperative: Why Architecture Matters

~400ms

Finality

Enables instant secondary market trading, eliminating counterparty settlement risk entirely.

<$0.001

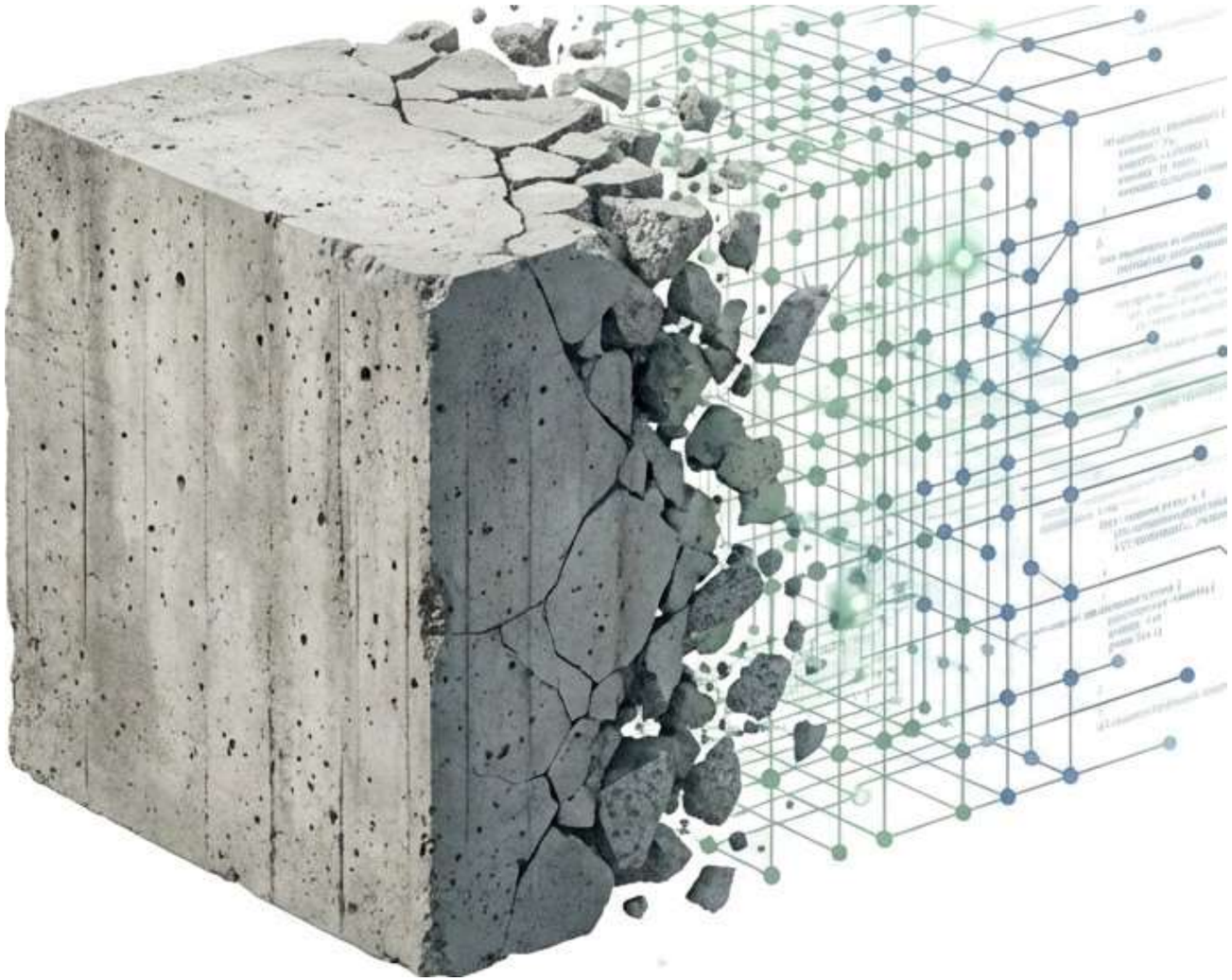
Avg Fee

The only way micro-distributions work. Distributing a $2 monthly rental yield is economically impossible if network gas fees are $5.

65,000+

TPS

Institutional-grade throughput capable of handling millions of fractional shares changing hands concurrently without network degradation.



The Internet of Value Has Arrived

We are bridging the gap between the ultra-stable, income-generating world of physical property and the ultra-ehicient, borderless world of decentralized finance.

$330 Trillion in static, illiquid value is finally moving at the speed of light.